MAIS PAC

IPhO 02.08: Circuits, Capacitors and Inductors · Worked Solutions

Circuits, Capacitors and Inductors: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Charging through a resistor. Return →

(a) $\tau = RC = (2.0\times10^{3})(100\times10^{(-6)}) = 0.20$ {s}; $I_0 = \varepsilon/R = 12/2000 = 6.0$ {mA}.

(b) At $t = 0.30$ {s} $= 1.5\tau$,

$$q = C\varepsilon(1-e^{(-1.5)}) = (1.2\times10^{(-3)})(1-0.223) = 9.3\times10^{(-4)}\ \{C\},\ I = I_0e^{(-1.5)} = 1.3\ \{mA\}. \quad (1)$$

(c) $U = (q^{2})/(2C) = ((9.32\times10^{(-4)})^{2})/(2\times10^{(-4)}) = 4.3\times10^{(-3)}$ {J}. The battery has supplied $q\varepsilon = (9.32\times10^{(-4)})(12) = 1.1\times10^{(-2)}$ {J}, so $6.9\times10^{(-3)}$ {J} has been dissipated in the resistor. **Check:** 4.3+6.9 = 11.2 {mJ} — the whole supply, as energy conservation demands.

Solution 2 — Discharging through two paths. Return →

(a) $U = (1)/(2)CV^2 = (1)/(2)(470\times10^{-6})(24)^2 = 0.135$ {J}.

(b) $V = V_0e^{-t/\tau}$ with $\tau = (1.5\times10^3)(470\times10^{-6}) = 0.705$ {s}, so

$$t = \tau\ln ()/(){V_0}{V} = 0.705\ln 4 = 0.98 \text{ {s}}. \quad (2)$$

(c) The current is $I = -(V_0/R)e^{-t/\tau}$, so the power in the resistor is $(V_0^2/R)e^{-2t/\tau}$, and

$$\int_0^\infty (V_0^2)/(R)e^{-2t/\tau},dt = (V_0^2)/(R)\cdot(\tau)/(2) = (V_0^2)/(R)\cdot(RC)/(2) = (1)/(2)CV_0^2. \quad (3)$$

The resistance cancels: it sets how long the discharge takes and how large the current is, but not how much energy is delivered. **Check:** with a very small R the current is large and the time short, and the product is unchanged.

Solution 3 — An inductor switched on. Return →

(a) After a long time the inductor is just a wire (of resistance 5.0 Ω): $I_\infty = 6.0/5.0 = 1.2$ {A}.

(b) $\tau = L/R = 0.25/5.0 = 0.050$ {s}. At $t = 0.030$ {s} $= 0.60\tau$,

$$I = 1.2(1-e^{-0.6}) = 1.2(0.451) = 0.54 \text{ {A}}. \quad (4)$$

(c) $V_L = \varepsilon - IR = 6.0 - (0.541)(5.0) = 3.3$ {V}. At t=0 the current is zero, so $V_L = 6.0$ {V} — the whole supply. **Check:** $V_L = L,dI/dt = (0.25)(1.2/0.05)e^{-0.6} = 6.0(0.549) = 3.3$ {V}, matching. As $t\to\infty$, $dI/dt\to0$ and $V_L\to0$: the inductor becomes invisible.

Solution 4 — LC oscillation. Return →

(a) $\omega_0 = 1/\sqrt{(LC)} = 1/\sqrt{((20\times10^{-3})(2.0\times10^{-6}))} = 1/\sqrt{(4.0\times10^{-8})} = 5.0\times10^3$ {rad,s^(-1)}, so $f_0 = 796$ {Hz} and $T = 1.26$ {ms}.

(b) Energy conservation, $(1)/(2)LI_{max}^2 = (1)/(2)CV_0^2$:

$$I_{max} = V_0\sqrt{((C)/(L))} = 50\sqrt{((2.0\times10^{-6})/(20\times10^{-3}))} = 50(1.0\times10^{-2}) = 0.50 \text{ {A}}. \quad (5)$$

(c) $U_C = (1)/(2)(2.0\times10^{-6})(25)^2 = 6.3\times10^{-4}$ {J}, and $U_L = U_{\{total\}} - U_C = (1)/(2)(2.0\times10^{-6})(50)^2 - 6.3\times10^{-4} = 2.5\times10^{-3} - 6.3\times10^{-4} = 1.9\times10^{-3}$ {J}. **Check:** $(1)/(2)LI^2$ with $I = \sqrt{(2U_L/L)} = 0.43$ {A} gives the same value, and $U_C + U_L = 2.5$ {mJ} as at the start.

Solution 5 — A driven series RLC circuit. Return →

(a) f_0 = (1)/(2π√(LC)) = (1)/(2π√((50×10^(-3))(20×10^(-6)))) = (1)/(2π(1.0×10^(-3))) = 159 {Hz}.

(b) Z = R = 10 Ω at resonance, so I_({rms}) = 10/10 = 1.0 {A}.

(c) V_C = IX_C = (I)/(ω_0C) = (1.0)/((1000)(20×10^(-6))) = 50 {V} — five times the supply. The inductor has an equal voltage IX_L = 50 {V} at every instant exactly opposite in phase, so the two cancel in the loop equation while each is separately large. **Check:** $V_R^2 = V^2$ at resonance and indeed 10 {V} (rms) appears across the resistor alone.

Solution 6 — Quality factor and bandwidth. Return →

(a) Q = (1)/(R)√((L)/(C)) = (1)/(10)√((50×10^(-3))/(20×10^(-6))) = (1)/(10)(50) = 5.0, and Δ f = f_0/Q = 159/5.0 = 32 {Hz}.

(b) The current falls to 1/√(2) of its peak when |X_L - X_C| = R, so

$$f = f_0\left(\sqrt{1 + \frac{1}{4Q^2}} \pm \frac{1}{2Q}\right) \quad f \approx 159{,}(1 \pm 0.1)\ \{\mathrm{Hz}\}, \tag{6}$$

that is 143 {Hz} and 175 {Hz} — separated by the bandwidth, as expected.

(c) As R→0 the peak current V/R diverges and the peak gets infinitely narrow; the power delivered is $I^2R = V^2/R$, which also diverges. In practice the resistance of the coil winding, the internal resistance of the source and the radiation resistance of the circuit are always present, and the components have voltage ratings that a high-Q resonance easily exceeds. **Check:** QΔ f = f_0 by construction, and the two half-power frequencies are placed symmetrically about f_0 on a logarithmic scale to first order in 1/Q.

Solution 7 — A capacitor network. Return →

(a) C_(23) = 6.0 + 12 = 18 μ{F}, then (1)/(C) = (1)/(4.0) + (1)/(18), so C = 3.27 μ{F} ≈ 3.3 μ{F}.

(b) Q_({total}) = CV = (3.27×10^(-6))(24) = 7.85×10^(-5) {C}, so C_1 carries 78 μ{C} and V_1 = Q/C_1 = 19.6 {V}; the parallel pair has V_(23) = 24 - 19.6 = 4.4 {V}, giving Q_2 = 26 μ{C} and Q_3 = 52 μ{C}.

(c) U = (1)/(2)CV^2 = (1)/(2)(3.27×10^(-6))(576) = 9.4×10^(-4) {J}. All in parallel, C = 22 μ{F} and U = (1)/(2)(22×10^(-6))(576) = 6.3×10^(-3) {J} — nearly seven times more for the same supply. **Check:** Q_2 + Q_3 = 78 μ{C} = Q_1, as charge conservation at the junction requires.

Solution 8 — Two capacitors sharing charge. Return →

(a) Charge is conserved: Q = (10×10^(-6))(100) = 1.0×10^(-3) {C}, and C = 30 μ{F}, so V = Q/C = 33 {V}. Then q_1 = 3.3×10^(-4) {C} and q_2 = 6.7×10^(-4) {C}.

(b) Initially U_i = (1)/(2)(10×10^(-6))(100)² = 5.0×10^(-2) {J}; finally U_f = (1)/(2)(30×10^(-6))(33.3)² = 1.7×10^(-2) {J}. The difference, 3.3×10^(-2) {J}, is dissipated in the connecting wire (and radiated).

(c) For two capacitors sharing charge, the loss is always

$$\Delta U = (1)/(2)(C_1C_2)/(C_1+C_2)(V_1-V_2)^2 = (1)/(2)(6.67\times10^{-6})(100)^2 = 3.3\times10^{-2}\ \{J\}, \tag{7}$$

which depends only on the initial and final states, not on R. The current that flows is set by R, but the total heat is ∫ I²R,dt: as R shrinks the current grows while the duration shrinks, leaving the integral unchanged. **Check:** with the numbers, (1)/(2)(10)(20)/(30) μ{F}×(100 {V})² = 33 {mJ}, matching the difference U_i-U_f = (5.0-1.7)×10^(-2) {J}.

Solution 9 — An RC timing circuit. Return →

(a) τ = RC = (50×10³)(22×10^(-6)) = 1.1 {s}, and V = ε(1-e^(-t/τ)) gives

$$1 - e^{(-t/1.1)} = (4.0)/(9.0) = 0.444\ e^{(-t/1.1)} = 0.556\ t = 1.1\ln(1.80) = 0.65\ \{s\}. \tag{8}$$

(b) I = (ε - V)/(R) = (9.0 - 4.0)/(5.0×10^4) = 1.0×10^(-4) {A} = 0.10 {mA}.

(c) Doubling C doubles τ, and the delay becomes 2×0.65 = 1.3 {s}, because the fraction of the final voltage reached is unchanged. The delay is set by the product RC and is insensitive to the exact supply voltage only if the threshold scales with ε; that is the weakness of the simple RC timer, and the reason precision timers use a comparator with a fixed reference. **Check:** e^(-0.65/1.1) = 0.556 and 1-0.556 = 0.444, giving 4.0 {V} exactly.

Solution 10 — A capacitor, a switch and two resistors. Return →

(a) After a long time the capacitor carries no current, so R_1 and R_2 form a simple divider:

$$V_C = \varepsilon(R_2)/(R_1+R_2) = 12\times(60)/(90) = 8.0\ \{V\},\ I = (12)/(90\times10^{3}) = 0.13\ \{mA\}. \tag{9}$$

(b) While the capacitor charges, the battery sees R_1 in series with R_2 C, so the charging time constant uses the resistance **seen by the capacitor**, which is R_2 in parallel with R_1 (the battery is a short circuit as far as the capacitor is concerned):

$$R_(\{eff\}) = (R_1R_2)/(R_1+R_2) = ((30)(60))/(90)\ \{k\Omega\} = 20\ \{k\Omega\},\ \tau = R_(\{eff\})C = (2.0\times10\text{^}4)(5.0\times10\text{^}(-6)) = 0.10\ \{s\}. \tag{10}$$

(c) The capacitor heads for 8.0 {V} with this time constant, so

$$6.0 = 8.0(1-e\text{^}(-t/0.10))\ e\text{^}(-t/0.10) = 0.25\ t = 0.10\ln 4 = 0.14\ \{s\}. \tag{11}$$

The relevant resistance is the **Thévenin** resistance looking out of the capacitor's terminals, not the total loop resistance: the battery is an ideal source and short-circuits the path between the two ends of the divider, so R_1 and R_2 appear in parallel from the capacitor's point of view. **Check:** the time constant is shorter than R_1C = 0.15 {s} and shorter than R_2C = 0.30 {s}, as a parallel combination must be.